# Lin & Nonlinear Optimization

Apr 1, 2026 Gst.

We can now relate second derivatives & convexity.

<u>Theorem:</u> Let $f \in C^2(\Omega)$. Then $f$ is convex on $\Omega$ if and only if $\nabla^2 f(\underline{x})$ is sym. pos. semidefinite for <u>all</u> $\underline{x} \in \Omega$.

<u>Proof:</u> "$\Longrightarrow$": Taylor theorem with exact remainder

$$f(\underline{y}) = f(\underline{x}) + \nabla f(\underline{x})^T(\underline{y}-\underline{x}) + \tfrac{1}{2}(\underline{y}-\underline{x})^T \nabla^2 f(\underline{\xi})(\underline{y}-\underline{x})$$

$\underline{\xi}$ unknown

If $\nabla^2 f(\underline{\xi})$ spd, then clearly

$$f(y) \geq f(x) + \nabla f(\underline{x})^T(\underline{y}-\underline{x}) \Longrightarrow f \text{ convex due to previous prop.}$$

"$\Longleftarrow$" See book. $\square$.

<u>Thus:</u> Local second-order condition equivalent to convexity close to minimizer.

<u>Examples:</u>
- $f(\underline{x}) = x^2$ and $f(x) = x^4$ are convex.
- $f(x) = \frac{1}{x}$ on $(0,\infty)$
- $f(\underline{x}) = \frac{1}{2}\underline{x}^T A \underline{x} + b^T\underline{x}$, $b \in \mathbb{R}^n$, $A \in \mathbb{R}^n$ is convex if $A$ is pos. semidefinite, since $\nabla^2 f(x) = A \in \mathbb{R}^{n\times n}$.
- $f(x) = e^x$

<u>Thm:</u> Let $f$ be convex on $\Omega \subset \mathbb{R}^n$. Then, every local minimum of $f$ is global.

<u>Proof:</u> $\underline{x}^* \in \Omega$ local minimum. Say $\underline{y} \in \Omega$ has $f(\underline{y}) < f(\underline{x}^*)$.

Then $f(\alpha y + (1-\alpha) x^*) \leq \alpha f(y) + (1-\alpha) f(x^*) < f(x^*)$,
which contradicts that $\underline{x}^*$ is local minimum.

Now show that for $f$ convex, the first-order necessary conditions are sufficient.

Thm: $f \in C^1$. If $\underline{x}^* \in \Omega$ satisfies

$$\nabla f(x^*)(\underline{y}-\underline{x}^*) \geq 0 \quad \text{for all } y \in \Omega,$$

then $x^*$ is a global minimum.

Proof: $y - x^*$ is feasible direction due to convexity of $\Omega$.
Gradient inequality:

$$f(y) \geq f(x^*) + \nabla f(\underline{x}^*)^T (\underline{y}-\underline{x}^*) \geq f(x^*). \quad \square.$$

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Algorithms: We will study iterative descent algorithms (gradient-based algorithms) of the following form:

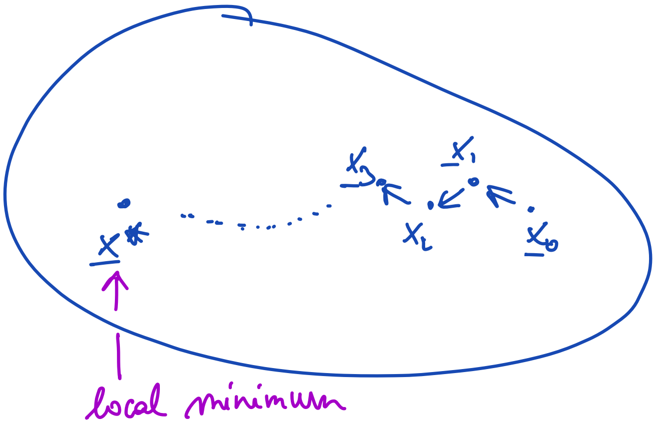
$x_0 \in \mathbb{R}^n$ ← initialization, start of algorithm

$$\begin{cases} x_{k+1} = \mathcal{A}(x_k) \\ f(x_{k+1}) < f(x_k) \\ \lim \underline{x_k} \to \underline{x}^* \end{cases}$$

$x_{k+1} = \mathcal{A}(x_k)$ ← "Some method to generate a new iterative point from old one"

$f(x_{k+1}) < f(x_k)$ ← "descent"

$\underline{x}^*$ ← local minimum




We need to answer:

- How should such an algorithm look like? When does it converge, and how does it depend on the initialization $\underline{x}_0$?
- How fast does it converge in term of # of iterations?
- How can we understand the tradeoffs between different methods?

## Speed of convergence

**Def:** We say a sequence of real numbers $\{\mu_k\}_{k\geq 0}^{\infty}$ converges with order $p$ if

$$\lim_{k\to\infty} \frac{|\mu_{k+1} - \mu^*|}{|\mu_k - \mu^*|^p} = \mu < \infty.$$

This is determined by the tail behavior of the sequence, not early stages.

If $\mu_k \to \mu^*$ with order $p$, then

$$|\mu_{k+1} - \mu^*| = \mu \cdot |\mu_k - \mu^*|^p,$$ so higher $p$ is faster convergence (typically we care about $p = 1, 2$)

**Ex I:** $\mu_k = a^k$ with $0 < a < 1$, $\mu^* = 0$

$$\frac{|\mu_{k+1}-0|}{|\mu_k-0|} = a \Rightarrow p=1$$

Ex II: $\mu_k = a^{2^k}$, $0<a<1$

$$\frac{\mu_{k+1}}{\mu_k^p} = a^{2^{k+1}-p\cdot 2^k} = a^{2^k(2-p)}$$

Thus, for $p=2$, $\frac{\mu_{k+1}}{\mu_k^2} \longrightarrow 1$, thus second-order (quadratic) convergence.

For $p=1$, we need a refined definition:

We say that $\{\mu_k\}$ converges to $\mu^*$ linearly with rate $\beta<1$

if $$\lim_{k\to\infty} \frac{|\mu_{k+1}-\mu^*|}{|\mu_k-\mu^*|} = \beta < 1 \quad (\text{requires } \beta<1)$$

Asymptotically: $|\mu_{k+1}-\mu^*| \leq \beta\, |\mu_k-\mu^*|$

e.g. with $\beta=\frac{1}{2}$, the distance roughly halves each step.

Arithmetic Convergence: $\{\mu_k\}$ converges to $\mu^*$ arithmetically with order $p$ if

$$|\mu_k-\mu^*| \leq C\,\frac{|\mu_0-\mu^*|}{k^p} \qquad k\geq 1,\ 0<p<\infty,\ C>0.$$

higher $p$, faster convergence

Ex III: $\mu_k = \frac{1}{k}$, $\mu^* = 0$

$$\lim_{k\to\infty} \frac{|\mu_{k+1}|}{|\mu_k|} = 1 \quad (\text{not linear!})$$

arithmetic with order 1.

Convergence speed comparison:

order $p$, $p>1$ (e.g. quadratic) $\quad$ linear $\quad$ arithmetical, any $p$

faster $\longleftrightarrow$ slower